

AKLT Model Analysis

This notebook analyzes the AKLT (Affleck-Kennedy-Lieb-Tasaki) model using matrix product states and explores its symmetry properties. Below is the code which finds the symmetry group's underlying Lie algebra's generator in both the physical level and the virtual level.

Import Required Libraries

```
In [1]: from emachine_v2 import aklt, pprint
import numpy as np
from matplotlib import pyplot as plt
from SUN_Centraliser import TensorProductGroup as TPG
```

Initialize AKLT MPS and Parent Hamiltonian

The AKLT state was initialised with the built-in `aklt()` function of the `emachine_v2.py` script. The chain was also generated with a boundary matrix `np.array([[1,0],[0,1]])`, which is under the PBC condition. The parent hamiltonian is found using orthogonal construction of the MPS. The hamiltonian in consideration have local projectors of length 2, and total length 3.

```
In [2]: # Initialize AKLT matrix product state
mps = aklt()
bound = np.array([[1,0],[0,1]])

# Compute parent Hamiltonian with range 2, bond dimension 3
ph_2_3 = mps.parent_hamiltonian(2,3)
```

Ground State Computation

```
In [3]: # Convert to ground space
state = mps.to_ground_space(3, bound)
```

SUN Symmetry Analysis

The symmetry of the AKLT model was assumed to be continuous and belongs to some group isomorphic to the $SU(3)$ group on the physical level. A basis of the $\mathfrak{su}(3)$ Lie algebra generators are found using a construction of the Gell-Mann Basis. As the symmetry group G of the MPS is a group which commutes with the Parent Hamiltonian,

we can construct the symmetry group by constructing the linear operator of the map $F := [-, H] : \mathfrak{g} \rightarrow \mathcal{M}^{d' \times d'}$. The symmetry group is then generated by the kernel of F .

```
In [4]: # Extract SUN symmetry generators
U1, U2, U3 = mps.sun_symmetry(ph_2_3, return_coeff=False, full_output=True)

# Apply generator transformation
umps = mps.apply_generator(U1 + 2*U2, theta=np.pi/2)

# Compute transformed ground state
ustate = umps.to_ground_space(3, bound)
```

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SYMMETRIC GENERATORS ANALYSIS
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```

Found 3 generator(s) in the null space:

Generator 1:

Null space basis vector index: 0
Coefficients in local su(3) basis: [-0.6445+0.j 0. +0.j -0.6445+0.j
0.0094-0.2904j
0. +0.j 0.0094-0.2904j 0.0105+0.0034j 0.0182+0.0059j]

Basis generators in su(3) (with non-zero coefficients):

[0] (coeff: (-0.6445+0j)):
[[0.+0.j 1.+0.j 0.+0.j]
[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: (-0.6445+0j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: (0.0094-0.2904j)):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: (0.0094-0.2904j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: (0.0105+0.0034j)):
[[0.7071+0.j 0. +0.j 0. +0.j]
[0. +0.j -0.7071+0.j 0. +0.j]
[0. +0.j 0. +0.j 0. +0.j]]
[7] (coeff: (0.0182+0.0059j)):
[[0.4082+0.j 0. +0.j 0. +0.j]
[0. +0.j 0.4082+0.j 0. +0.j]
[0. +0.j 0. +0.j -0.8165+0.j]]

Local su(3) generator:
[[0.0149+0.0048j -0.9349-0.0094j 0. +0.j]
[-0.354 +0.0094j 0. +0.j -0.9349-0.0094j]
[0. +0.j -0.354 +0.0094j -0.0149-0.0048j]]

Generator 2:

Null space basis vector index: 1
Coefficients in local su(3) basis: [0.2905+0.j 0. +0.j 0.2905+0.j
0.0202-0.6442j
0. +0.j 0.0202-0.6442j -0.0053+0.0081j -0.0092+0.0141j]

Basis generators in su(3) (with non-zero coefficients):

[0] (coeff: (0.2905+0j)):
[[0.+0.j 1.+0.j 0.+0.j]

```

[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: (0.2905+0j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: (0.0202-0.6442j)):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: (0.0202-0.6442j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: (-0.0053+0.0081j)):
[[ 0.7071+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j -0.7071+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j 0.      +0.j]]
[7] (coeff: (-0.0092+0.0141j)):
[[ 0.4082+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j 0.4082+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j -0.8165+0.j]]

```

Local su(3) generator:

```

[[-0.0075+0.0115j -0.3537-0.0202j 0.      +0.j      ]
[ 0.9347+0.0202j 0.      +0.j      -0.3537-0.0202j]
[ 0.      +0.j      0.9347+0.0202j 0.0075-0.0115j]]

```

Generator 3:

Null space basis vector index: 2

```

Coefficients in local su(3) basis: [-0.0167+0.j      0.      +0.j      -0.016
7+0.j      -0.0125+0.0006j
0.      +0.j      -0.0125+0.0006j -0.4997+0.0106j -0.8655+0.0184j]

```

Basis generators in su(3) (with non-zero coefficients):

```

[0] (coeff: (-0.0167+0j)):
[[0.+0.j 1.+0.j 0.+0.j]
[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: (-0.0167+0j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: (-0.0125+0.0006j)):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: (-0.0125+0.0006j)):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: (-0.4997+0.0106j)):
[[ 0.7071+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j -0.7071+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j 0.      +0.j]]
[7] (coeff: (-0.8655+0.0184j)):

```

```
[[ 0.4082+0.j  0.      +0.j  0.      +0.j]
 [ 0.      +0.j  0.4082+0.j  0.      +0.j]
 [ 0.      +0.j  0.      +0.j -0.8165+0.j]]
```

```
Local su(3) generator:
[[-0.7066+0.015j -0.0161+0.0125j  0.      +0.j      ]
 [-0.0172-0.0125j  0.      +0.j      -0.0161+0.0125j]
 [ 0.      +0.j      -0.0172-0.0125j  0.7066-0.015j ]]
```

Null space basis (3 vectors):

```
Basis vector 1 (norm = 1.0000):
[-0.6445+0.j      0.      +0.j      -0.6445+0.j      0.0094-0.2904j
 0.      +0.j      0.0094-0.2904j  0.0105+0.0034j  0.0182+0.0059j]
```

```
Basis vector 2 (norm = 1.0000):
[ 0.2905+0.j      0.      +0.j      0.2905+0.j      0.0202-0.6442j
 0.      +0.j      0.0202-0.6442j -0.0053+0.0081j -0.0092+0.0141j]
```

```
Basis vector 3 (norm = 1.0000):
[-0.0167+0.j      0.      +0.j      -0.0167+0.j      -0.0125+0.0006j
 0.      +0.j      -0.0125+0.0006j -0.4997+0.0106j -0.8655+0.0184j]
```

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Alternative Parent Hamiltonian

```
In [5]: # Compute parent Hamiltonian with range 3, bond dimension 3
ph_3_3 = mps.parent_hamiltonian(3,3)
#mps.sun_symmetry(ph_3_3, return_coeff=False)
```

Virtual Symmetry Analysis

```
In [6]: # Extract virtual symmetry generators
V1, V2, V3 = [v["generator"] for v in mps.virtual_symmetry(ph_2_3, verbose=1)]

print(V1, '\n', V2, '\n', V3)
```

```
Physical generator G:
[[ 0.0149+0.0048j -0.9349-0.0094j  0.      +0.j      ]
 [-0.354 +0.0094j  0.      +0.j      -0.9349-0.0094j]
 [ 0.      +0.j      -0.354 +0.0094j -0.0149-0.0048j]]
```

```
Virtual generator found:
Phase: -0.000000+0.000000j
Error: 1.350398e-15
Relative error: 1.169622e-15
Generator K:
[[-0.0074-0.0024j  0.2503-0.0067j]
 [ 0.6611+0.0067j  0.0074+0.0024j]]
```

```
Physical generator G:
[[-0.0075+0.0115j -0.3537-0.0202j  0.      +0.j      ]
 [ 0.9347+0.0202j  0.      +0.j      -0.3537-0.0202j]
 [ 0.      +0.j      0.9347+0.0202j  0.0075-0.0115j]]
```

```
Virtual generator found:
Phase: 0.000000+0.000000j
Error: 2.357900e-15
Relative error: 2.042194e-15
Generator K:
[[ 0.0037-0.0058j -0.6609-0.0143j]
 [ 0.2501+0.0143j -0.0037+0.0058j]]
```

```
Physical generator G:
[[-0.7066+0.015j -0.0161+0.0125j  0.      +0.j      ]
 [-0.0172-0.0125j  0.      +0.j      -0.0161+0.0125j]
 [ 0.      +0.j      -0.0172-0.0125j  0.7066-0.015j ]]
```

```
Virtual generator found:
Phase: -0.000000-0.000000j
Error: 7.386672e-16
Relative error: 9.042873e-16
Generator K:
[[ 0.3533-0.0075j  0.0122+0.0088j]
 [ 0.0114-0.0088j -0.3533+0.0075j]]
[[-0.00743927-0.00239986j  0.25034121-0.00668117j]
 [ 0.66105642+0.00668117j  0.00743927+0.00239986j]]
[[ 0.00374965-0.00575473j -0.66094876-0.01431553j]
 [ 0.25009629+0.01431553j -0.00374965+0.00575473j]]
[[ 0.35332019-0.00752106j  0.01219453+0.00881653j]
 [ 0.01135546-0.00881653j -0.35332019+0.00752106j]]
```

Jacobi Identity Verification

```
In [7]: # Verify Jacobi identity for the virtual symmetry generators
pprint(TPG.jacobi_identity(V1, V2, V3))
```

```
[[0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j]]
```